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Evaluating the Potential of Using Free Remote Sensing Data for Automated Agricultural Mapping: A Comparative Study Between KNN and Random Forest Algorithms as a Tool to Support Land Resource Management.

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Abstract

Accurate and up-to-date mapping of agricultural crops constitutes critical infrastructure for land resource management and environmental planning. However, the high dynamism of agricultural areas and the phenological similarity between different crops pose a significant challenge in automated satellite imagery-based identification. This study examines the potential of integrating time series of the NDVI index from free satellite data (Sentinel-2) with machine learning algorithms, for the spatial classification of crops in the Yoav Regional Council. The research was based on a comparative examination between a tree-based ensemble model (Random Forest) and a distance-based model (K-Nearest Neighbor).

The research findings indicate the clear superiority of the Random Forest model, which presented a higher overall accuracy (51.2% compared to 46.1%) and produced homogeneous classification maps that faithfully reflect the agronomic reality on the ground, while filtering out "noise" and reducing the pixelation effect. In addition, the analysis shows that both models exhibited very high performance (over 71% accuracy for RF) in identifying evergreen orchards (avocado and olive) , but struggled to separate seasonal field crops (such as wheat and clover) due to almost complete phenological overlap.

The study concludes that integrating free Sentinel-2 data and the RF algorithm constitutes a practical, available, and economical tool for environmental and agricultural monitoring. Furthermore, the findings pave the way for focused follow-up studies that will incorporate Object-Based Image Analysis (OBIA) and supplementary structural data (such as Sentinel-1 SAR radar and texture analysis) to overcome the barriers of spectral separation and improve mapping accuracies.

1. Introduction

Accurate and updated mapping of land uses, and particularly of agricultural crop types, constitutes critical information infrastructure for natural resource management, environmental planning, and ecological risk assessment. While traditional field surveys or manual decoding provide data with a high level of reliability, these processes are resource-intensive, time-consuming, and difficult to implement or update frequently at the regional level. Remote Sensing technologies and Geographic Information Systems (GIS) offer an efficient, economical, and continuous technological alternative for monitoring the earth's surface on a large scale, while reducing the dependence on physical fieldwork.

Unlike static land covers such as built-up areas or infrastructure, agricultural areas are characterized by extremely high dynamism. This dynamism stems from the natural cyclicality of vegetation (phenology) and proactive agricultural practices such as crop rotation. As a result, identifying the crop type based on a single-date satellite image is often insufficient. Different crops may display an identical Spectral Signature at certain developmental stages, for example, when both fields are at their peak green bloom or after harvest. Therefore, current research literature points to the essential need for Multi-temporal Analysis. This analysis tracks changes in vegetation indices, such as the NDVI (Normalized Difference Vegetation Index), calculated by the ratio between the NIR and Red bands:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Continuous tracking throughout the entire growing season allows for the construction of a unique "phenological-spectral signature" for each crop type.

The availability of data from satellites with high spatial and temporal resolution, such as the European Space Agency's (ESA) Sentinel-2 satellite constellation, has opened new avenues in agricultural monitoring capabilities. These sensors provide imagery at a frequency of about five days, allowing close monitoring of growth stages. However, dealing with enormous volumes of information (Big Data) requires shifting from basic classification methods to using advanced Machine Learning algorithms. This study focuses on comparing two main classification approaches: the **Random Forest (RF)** algorithm, based on an ensemble of decision trees, and the **K-Nearest Neighbors (KNN)** algorithm, based on similarity and distance in the spectral feature space. The ability of these models to identify hidden patterns within spatial data allows for highly accurate automatic pixel classification, based on Ground Truth data that train the model.

In the Israeli space, and particularly in the Yoav Regional Council, the challenge of automated classification intensifies due to the nature of local agriculture: intensive, highly diverse agriculture spread over relatively small land units. Developing a reliable classification model in this complex environment is a powerful tool. Beyond the mapping capability itself, building a model capable of identifying crops remotely is a central pillar in analyzing the population's potential exposure to specific pesticides, which are directly derived from the type of crop located near their residences.

Research Objectives

The overarching goal of this research is to develop, implement, and validate a machine learning-based methodology for the automated identification and classification of agricultural crop types in the Yoav Regional Council area, using free satellite imagery from the Sentinel-2 network.

The specific objectives include:

- **Comparing classification performance:** Examining the accuracy level of the Random Forest algorithm versus the KNN algorithm in mapping 5 selected crop types.
- **Multi-temporal analysis:** Examining the contribution of NDVI time series to the separation capability between crops with a similar spectral signature throughout the growing season.
- **Accuracy Assessment:** Validating the spatial results using a Confusion Matrix and Overall Accuracy and Kappa metrics, based on ground truth data.

2. Methods

2.1 Study Area and Data Sources

The research focuses on the agricultural space of the Yoav Regional Council, an area characterized by intensive, diverse, and frequently changing agricultural activity, which serves as an ideal test environment for developing classification models. The spatial database for the study is based on the integration between remote sensing data and administrative vector data:

- **Satellite Imagery:** The spatial analysis is based on Time-Series of imagery from the Sentinel-2 satellite belonging to the European Space Agency (ESA). The imagery was downloaded at the atmospherically corrected processing level (Level 2A - Bottom of Atmosphere) at a spatial resolution of 10 meters. In order to monitor the phenology (growth cycle) of the different crops, images from several key dates throughout the agricultural growing season were selected. A strict filtering was performed for images with a cloud cover lower than 10%, to prevent surface masking and minimize errors in spectral reflectance.
- **Ground Truth:** The Target Variable was derived from a public agricultural plots vector polygon layer (Shapefile) maintained by the Ministry of Agriculture. These data underwent preliminary filtering so that only plots whose "update date" value exactly matched the year of the satellite imagery were included. This process is essential to ensure research traceability and to prevent statistical biases resulting from farmers' crop rotation practices.

2.2 Spatial Pre-processing

The initial processing chain and Feature Engineering were performed in the ArcGIS Pro software environment. First, all satellite images were geometrically cut to the study area boundaries using a Clip tool, to streamline calculation processes.

For each monthly timestamp, the Normalized Difference Vegetation Index (NDVI) was calculated, which is a quantitative and stable indicator of the vegetation's growth level and photosynthetic activity. The index is calculated based on the normalized difference between the reflectance in the Near-Infrared (NIR) band and the visible Red band, according to the formula:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

The NDVI indices, along with the standard color bands (RGB) from all sampled months, were consolidated into a Multi-temporal Raster Stack using the Composite Bands tool. This methodological approach enables the construction of a spatio-temporal spectral signature for each pixel in the space, representing the development of the vegetation over time, instead of relying on a momentary and static signature.

2.3 Machine Learning Architecture and Classification Workflow

The classification of land cover and agricultural crops was carried out in a comparative approach, using two supervised machine learning algorithms: K-Nearest Neighbors (KNN) as a baseline model, and Random Forest (RF) as an advanced Ensemble model. While the KNN algorithm classifies each pixel based on statistical similarity (Euclidean distance) to the K pixels closest to it, the RF model is based on building a "forest" of random decision trees and weighting their predictions, a feature that gives it higher resistance to noise in the data space. The process was implemented in the ArcGIS Pro environment and included three sequential steps:

- **Training Samples Generation:** Using the Training Samples Manager panel, a hierarchical classification schema was built based on the "crop category" and "crop name" fields from the Ministry of Agriculture's Ground Truth polygons. From these polygons, representative training areas were sampled for each crop type. For each sample, a unique category name and numerical value were defined, which aggregated the pixel data from the multi-temporal rasters. Exactly the same training samples were used to feed both models.
- **Model Training:** The training was performed separately for each algorithm using the Train K-Nearest Neighbor Classifier and Train Random Forest Classifier tools. At this stage, the structural parameters of the models were defined: for the RF model, 100 decision trees (Number of Trees) were used to ensure statistical stability, and for the KNN model, a value of K=1 (Number of Neighbors) was set for classification based on the nearest neighbor in the feature space. "The models were fed with the multi-layered file that unifies the time data (which constitutes the set of explanatory variables) and the training polygons layer." At the end of the data processing, the system produced two trained model definition files (in .ecd format).
- **Raster Classification:** The trained models were applied to all pixels in the research space using the Classify Raster tool. The final product obtained is two classified rasters in Esri's Cloud Raster Format (CRF), which allow

spatial comparison between the predictions of the KNN model and those of the RF.

2.4 Accuracy Assessment and Quality Control

In order to validate the significance of the results and compare the models' performance objectively, a rigorous quality control system was built based on independent data that did not take part in the training phase:

- **Control Points Sampling:** Using the Create Accuracy Assessment Points tool, a layer of ground truth points was created within selected polygons from the Ground Truth data. The sampling strategy was implemented using the Equalized Stratified Random sampling method, allocating 100 points to each of the 5 crop categories (a total of 500 control points), to prevent statistical bias in favor of common crops in the field. The same control points were used to test both models.
- **Radiometric Intersection:** The control points layer was overlaid with each of the classified rasters, and the prediction values of the models were extracted into the attribute table of the points using the Update Accuracy Assessment Points tool.
- **Confusion Matrix:** The final accuracy analysis was quantified using the Compute Confusion Matrix tool. Two separate matrices were produced (one for KNN and one for RF) that provided a quantitative statistical comparison between the verified classification and the model's prediction. The accepted accuracy metrics were derived from the matrices: Overall Accuracy, Producer's Accuracy, and User's Accuracy. The significance of the statistical results was reported at an accuracy level of two significant digits, allowing the definition of the optimal algorithm for classification in this space.

3. Results

This chapter presents the spatial classification results of the agricultural crops in the Yoav Regional Council area, as obtained from the two machine learning models: Random Forest (RF) and K-Nearest Neighbors (KNN). The Accuracy Assessment was based on an independent sample of 494 valid control points.

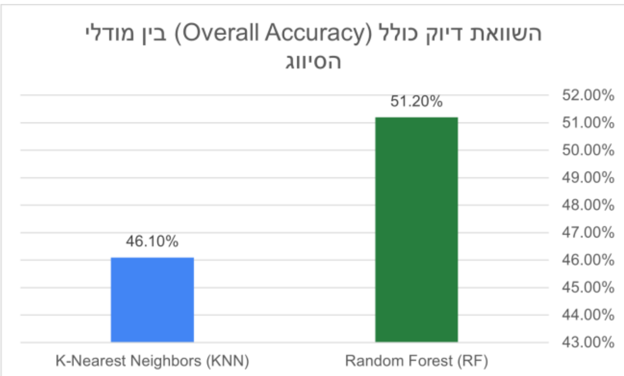
3.1 Overall Performance Comparison

Examination of the summary accuracy metrics indicates a gap in the performance of the two models, with the RF model demonstrating superior classification ability over the KNN model in all the examined metrics. Table 1 presents the Overall Accuracy and Kappa Coefficient values for each of the algorithms.

Table 1: Summary accuracy metrics for comparing the classification models.

דיוק כולל (Overall Accuracy)	מקדם קאפא (Kappa Coefficient)	מודל סיווג
51.20%	0.39	**Random Forest (RF)**
46.10%	0.327	**K-Nearest Neighbors (KNN)**

These data indicate that the use of a decision tree ensemble-based model (RF) enables better coping with the spatio-temporal spectral complexity of the agricultural area, compared to a distance-based model (KNN).



(Figure 1 Reference: Overall Performance Comparison between RF and KNN Models)

3.2 Confusion Matrices Analysis

For a focused examination of the classification quality at the level of the different crop categories, confusion matrices were produced for both models. Analyzing the matrices allows identifying the accuracy levels unique to each crop type (Producer’s and User’s Accuracy), as well as locating systematic error patterns (Misclassifications) between crops with similar spectral signatures.

	OBJECTID *	ClassValue	Avocado	Olive	Wheat	Corn	Clover	Total	U_Accuracy	Kappa
1	1	C_1	71	18	11	8	10	118	0.601695	0
2	2	C_2	15	71	17	12	18	133	0.533835	0
3	3	C_3	7	4	26	23	21	81	0.320988	0
4	4	C_4	2	4	14	49	14	83	0.590361	0
5	5	C_5	3	2	32	6	36	79	0.455696	0
6	6	Total	98	99	100	98	99	494	0	0
7	7	P_Accuracy	0.72449	0.717172	0.26	0.5	0.363636	0	0.512146	0
8	8	Kappa	0	0	0	0	0	0	0	0.390248

(Table 2 Reference: Confusion Matrix - Random Forest model)

	ClassValue	Avocado	Olive	Wheat	Corn	Clover	Total	U_Accuracy	Kappa
1	C_1	58	22	9	7	8	104	0.557	0
2	C_2	17	52	11	12	7	99	0.525	0
3	C_3	9	6	21	17	20	73	0.287	0
4	C_4	4	9	24	52	19	108	0.481	0
5	C_5	10	10	35	10	45	110	0.409	0
6	Total	98	99	100	98	99	494	0	0
7	P_Accuracy	0.591	0.525	0.21	0.531	0.455	0	0.461	0
8	Kappa	0	0	0	0	0	0	0	0.327

(Table 3 Reference: Confusion Matrix - K-Nearest Neighbors model)

From a specific examination of the KNN algorithm results, a considerable variance can be seen in the identification capability of different crops:

- **The highest accuracy level:** Was obtained for the avocado crop, which achieved a Producer’s Accuracy of 59.1%. This indicates that this crop is characterized by a relatively unique spectral signature, which the model managed to identify with the highest success rate. A similar trend was also recorded in the olive crop (52.5%), indicating the model's relative ability to identify orchards.

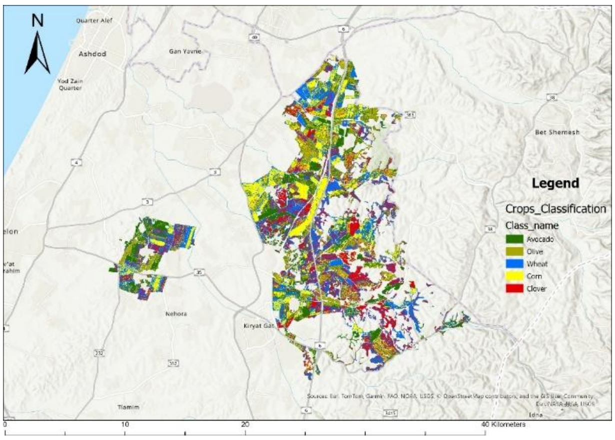
- **The lowest accuracy level and error patterns:** On the other hand, the model struggled significantly in identifying seasonal field crops, particularly wheat, which received a low producer accuracy of only 21.0%. From the analysis of the confusion matrix, it appears that most of the errors in this category stemmed from misclassification (confusion) with crops of the clover type (35 error points) and corn (24 error points). This result suggests a great similarity in the phenology and growth rate of the field crop group, making it difficult for the KNN algorithm to produce an accurate spatial separation between them.

Examining the Random Forest algorithm results indicates a similar trend in classification patterns, but with a significant improvement in overall performance, and particularly in the separation capability of orchards:

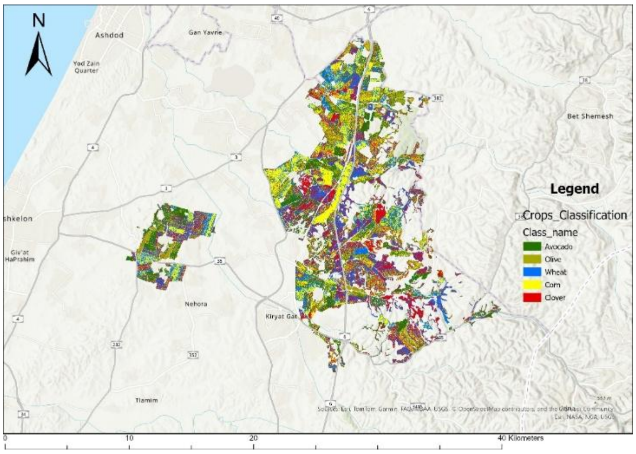
- **The leap in orchard accuracy:** Avocado and olive crops were identified at the highest accuracy level, which stood at a producer accuracy of 72.4% and 71.7% respectively. This is a dramatic improvement compared to the KNN model, a figure that emphasizes the clear advantage of tree-based algorithms in dealing with multi-temporal spectral signatures of evergreen orchards in the agricultural space.
- **The field crops challenge:** Despite the superiority of the RF model, the field crops group remained challenging. The wheat crop showed the lowest accuracy level (26.0%). Similar to KNN, most of the "confusion" occurred against clover (32 error points out of 100). This finding reinforces the conclusion that the limitation in identification stems from an inherent phenological-spectral similarity between these crops during growth stages, a difficulty that even an advanced ensemble algorithm struggles to fully resolve.

<u>3.3 Visual Representation of Spatial Classification</u>

Beyond the statistical metrics, a spatial examination of the classification maps provides important insights regarding the operation mode of the algorithms in the field.



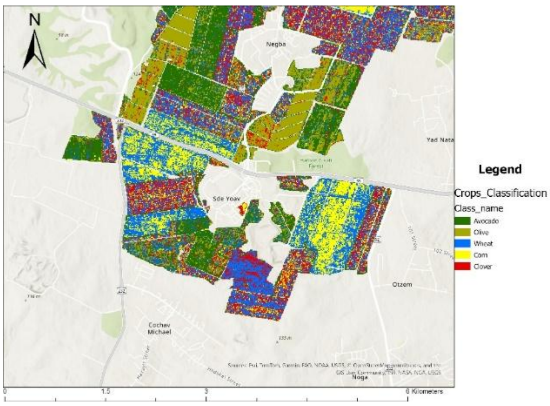
(Figure 2 Reference: Agricultural crops classification map as produced by the Random Forest model.)



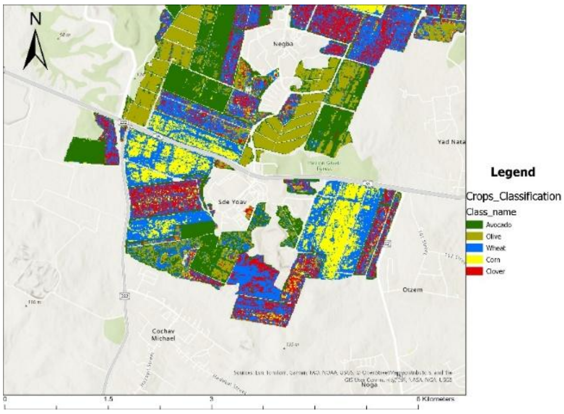
(Figure 3 Reference: Agricultural crops classification map as produced by the K-Nearest Neighbors model.)

A visual comparison between the classification maps demonstrates the spatial differences between the models. While the KNN model tends to produce a "dotted" and noisier output at the single pixel level, the Random Forest model manages to produce more homogeneous and continuous plots. This homogeneity more faithfully reflects the physical reality of managed commercial agricultural plots, reinforcing its preference as a mapping tool in the given space.

To illustrate the spatial differences between the model outputs, a focused local examination was performed in an area around Kibbutz Sde Yoav. A visual comparison of this area clearly demonstrates the superiority of the Random Forest model in representing ground reality.



(Figure 4 Reference: Agricultural crops classification map as produced by the K-Nearest Neighbors model, focusing on Kibbutz Sde Yoav and Kibbutz Negba.)



(Figure 5 Reference: Agricultural crops classification map as produced by the Random Forest model, focusing on Kibbutz Sde Yoav and Kibbutz Negba.)

In the prediction map of the KNN algorithm (first image), a very high noise level is evident at the single pixel level. It can be seen that within the boundaries of the same agricultural plot, pixels classified into a variety of different crops are mixed (for example, single pixels of wheat and corn scattered within clover fields or orchards). This fragmentation contradicts the agronomic logic of managing uniform commercial plots. In contrast, in the prediction map of the RF model (second image), the agricultural plots are obtained as homogeneous, continuous, and better-defined spaces than the KNN model managed to predict. The ability of the RF algorithm to weight multiple decision trees allowed it to filter out local spectral "background noises," and produce a continuous classification that faithfully reflects true agricultural patterns in the field.

4. Discussion and Conclusions

This study examined the ability to automatically classify agricultural crops in the Yoav Regional Council area, relying on free remote sensing data from the Sentinel-2 satellite and using machine learning algorithms. The study results clearly indicate the superiority of the Random Forest (RF) model over the K-Nearest Neighbors (KNN) model, both in the raw statistical metrics (Overall Accuracy and Kappa) and in the spatial-visual analysis of the classification products.

The Methodological Advantage of the Tree-Based Algorithm

The performance gap between the two models stems from their decision-making architecture. While the KNN algorithm classifies pixels based on proximity in the feature space (Euclidean distance), it struggles to deal with "noise" in the NDVI time series and tends to produce a fragmented classification (pixelated image). In contrast, the RF model, as an ensemble algorithm that examines subsets of variables through multiple decision trees, demonstrates higher immunity to spectral noises. This capability allowed the RF to identify the most critical variables for separation, and create classification maps with spatial homogeneity that faithfully reflects the agronomic reality of continuous plots.

The Effect of Phenology on Spectral Separation Capability

The confusion matrix analysis reveals a fascinating picture regarding the relationship between the crop type and its identifiability. Both models, and particularly the RF, demonstrated very high performance in identifying orchards (avocado and olive, over 71% accuracy for RF). This success is attributed to the fact that these are evergreen trees, which produce a relatively stable and high spectral signature throughout most months of the year, making it easier for the algorithm to isolate them from other crops.

On the other hand, the main challenge that remained without a complete response is the identification of seasonal field crops, led by wheat and clover. The high error rate between these two crops stems from almost complete phenological similarity: both are winter/spring crops that are sown, turn green, and dry out in parallel time windows. As a result, their NDVI curve throughout the growing season is almost completely identical. This finding proves that under conditions of tight phenological overlap, spectral and temporal resolution alone is insufficient, and it may be necessary in the future to incorporate radar (SAR) data that are sensitive to the physical structure of the plant and not just its greenness.

Summary and Applied Implications

The study proves that combining free Sentinel-2 data and the Random Forest algorithm constitutes a practical, available, and economical tool for agricultural monitoring, and presents advanced capabilities in dealing with a complex agricultural space. Beyond the mapping value itself, producing reliable information layers of crop distribution constitutes essential infrastructure for decision-making in the rural space and for more sustainable management of land and agricultural resources. At the same time, the study's findings and the challenges that arose from it - such as the difficulty in separating field crops with phenological overlap - emphasize that relying on spectral resolution at the single pixel level is not the whole picture. This insight paves the way for advanced follow-up studies that will focus on shifting from pixel-based classification to Object-Based Image Analysis (OBIA), as well as integrating supplementary structural data. Future development of these methodologies will enable maximizing the accuracy of crop maps, all while strictly maintaining the principle of using free information sources.

Future Research Directions: Transitioning from Pixel-Based Classification to Object-Based Image Analysis (OBIA)

A potential and significant development for follow-up studies is upgrading the methodology from single pixel-level classification (Pixel-based classification) to Object-Based Image Analysis (OBIA). While current models examine each spatial cell separately, an object-based model will first identify the boundaries of the agricultural plots as whole polygons. The research justification for this approach is based on the physical practice in the field: farmers sow and cultivate land cells as continuous and uniform blocks, not in random segments. This agronomic understanding makes an object-based algorithm particularly relevant and accurate for agriculture, as it adapts itself to spatial reality.

On top of the continuous polygons, a "Majority Voting" mechanism can be applied that analyzes the distribution of pixels within the plot. Thus, for example, a plot in which 70% of the pixels were identified as wheat and 30% as clover, will be fully classified as a homogeneous plot of wheat, effectively neutralizing local spectral "noises". In addition, it is recommended to integrate a "Confidence Threshold" based quality control mechanism in future models. In this mechanism, a plot that does not cross a pre-defined dominance threshold of pixels of a certain type will not be incorrectly classified, but rather marked under the "Unclassified" category. Implementing a strict threshold value will significantly reduce the False Positives rate and ensure that the final crop maps are of a high enough reliability level for regulatory and environmental planning purposes.

The Challenge of Classifying Evergreen Orchards and Integrating Free Structural Data (SAR and Texture Analysis)

Although the models in this study (and particularly the Random Forest) presented very high performance in separating avocado and olive trees from the rest of the field crops, it is important to note that this success relies largely on the distinct difference between the spectral signature of a fruit tree orchard and that of a standard seasonal crop. However, a significant challenge in follow-up studies will be the ability to differentiate between avocado trees and other evergreen crops with similar spectral signatures and foliage, such as citrus groves. Since these crops share almost identical chlorophyll density levels and phenology, relying on spectral resolution alone (such as the NDVI index) may prove insufficient and lead to misclassification.

To overcome this limitation while maintaining the methodological principle of using available and free remote sensing data, it is recommended to integrate variables based on the physical structure of the plot in future models. Every type of orchard is characterized by a "Spatial Roughness" and a unique landscape configuration. This structure can be quantified at no cost using two complementary approaches: first, applying Texture Analysis algorithms - such as the GLCM (Gray Level Co-occurrence Matrix) index - directly on the existing Sentinel-2 imagery; and second, adding free radar (SAR) data from the Sentinel-1 satellite. Radar waves are sensitive to the geometric structure and water content of the vegetation, and can provide the learning model with a physical-structural dimension in addition to the spectral dimension. The addition of these structural variables is expected to significantly increase the separation capability between different types of orchards, without increasing the costs of continuous monitoring.